



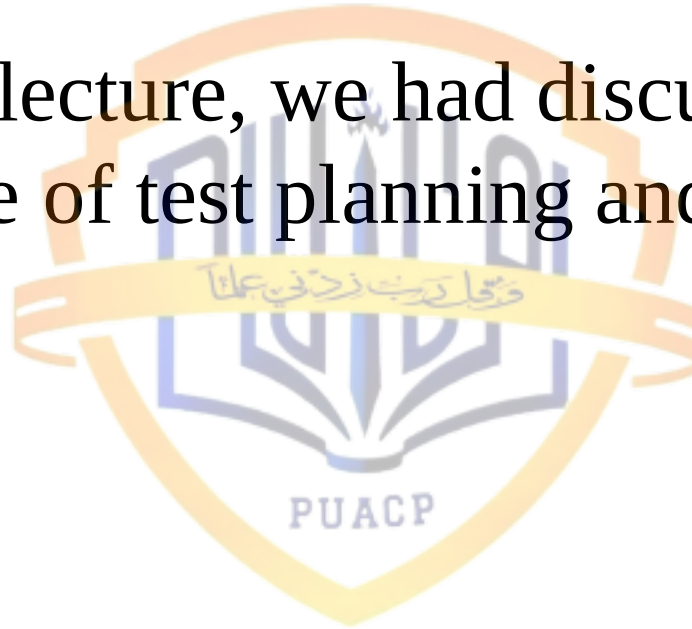
Software Testing – An Introduction to Automated Software Testing



Lecture # 31

Recap

- In the last lecture, we had discussed the importance of test planning and



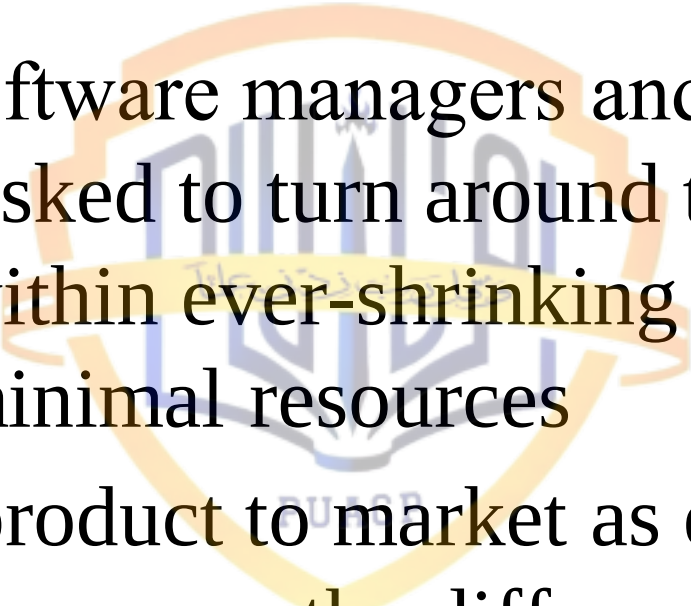
Today's Lecture

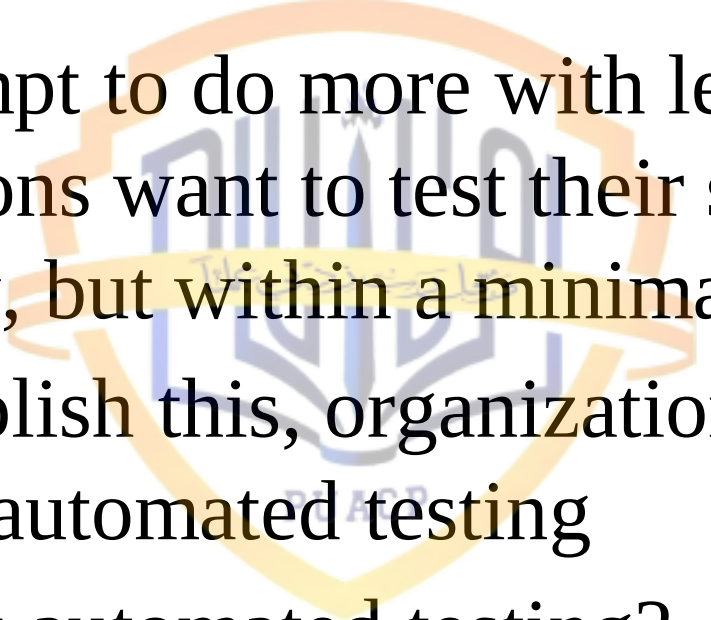
- Automated software testing



Introduction

- Studies have shown that companies are missing shipping deadlines for their software products
- Overall 90% of developers have missed ship dates, and missing deadlines is a routine occurrence for 67% of developers
- These numbers are very alarming for our industry

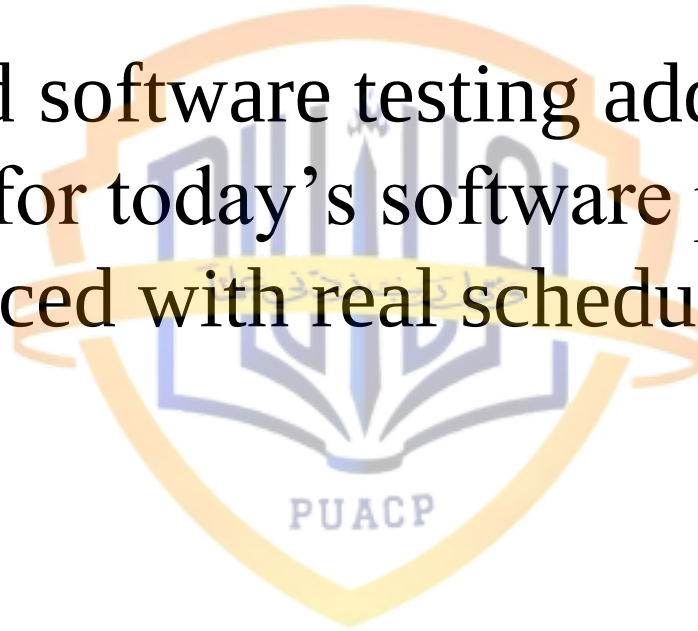
- 
- Today's software managers and developers are being asked to turn around their products within ever-shrinking schedules and with minimal resources
 - Getting a product to market as early as possible may mean the difference between product survival and product death – and therefore company survival and death

- 
- In an attempt to do more with less, organizations want to test their software adequately, but within a minimal schedule
 - To accomplish this, organizations are turning to automated testing
 - So, what is automated testing?

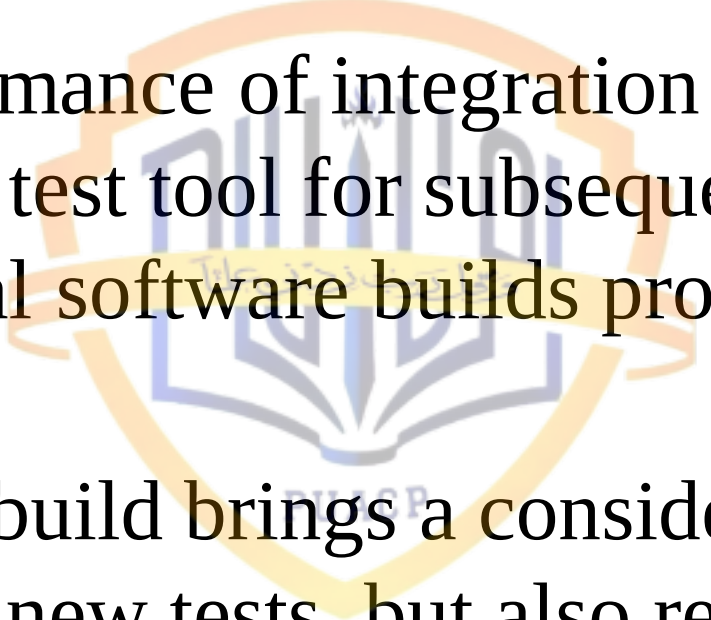
Automated Testing

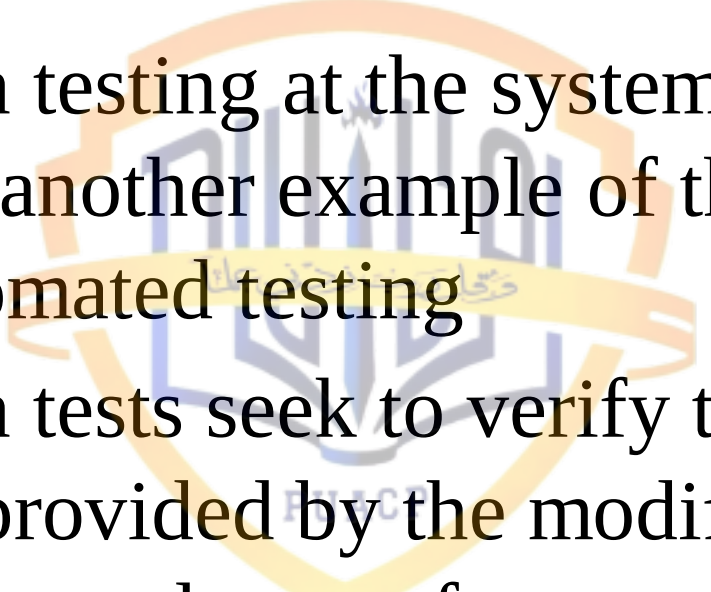
- The management and performance of test activities, to include the development and execution of test scripts so as to verify test requirements, using an automated test tool

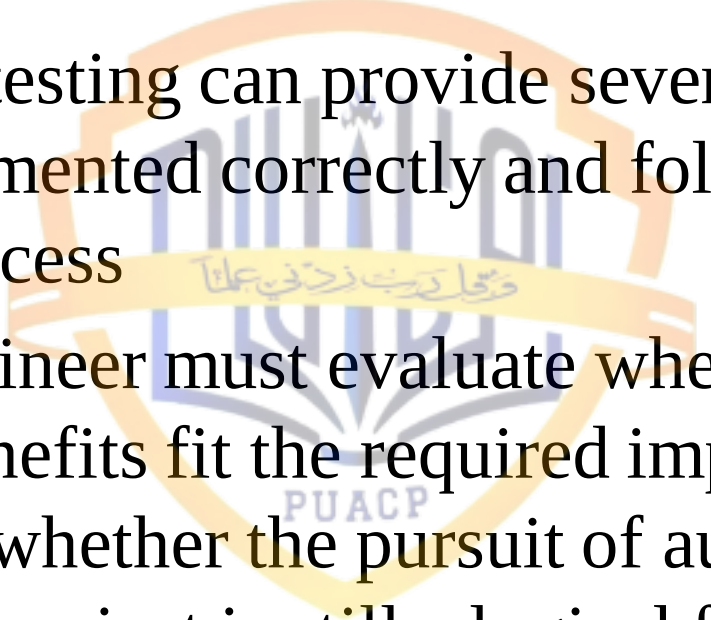
- Automated software testing addresses the challenge for today's software professionals who are faced with real schedule deadlines



- The automation of test activities provides its greatest value in instances where test scripts are repeated or where test script sub-routines are created and then invoked repeatedly by a number of test scripts

- 
- The logo of the University of Jordan is centered in the background. It features a shield with Arabic calligraphy, a book, and a sunburst. The text 'UNIVERSITY OF JORDAN' is written in English and Arabic around the shield.
- The performance of integration test using an automated test tool for subsequent incremental software builds provide great value
 - Each new build brings a considerable number of new tests, but also reuses previously developed test scripts

- 
- Regression testing at the system test level represents another example of the efficient use of automated testing
 - Regression tests seek to verify that the functions provided by the modified system or software product perform as specified and that no unintended change has occurred in the operation of the system or product

- 
- Automated testing can provide several benefits when implemented correctly and follows a rigorous process
 - The test engineer must evaluate whether the potential benefits fit the required improvement criteria and whether the pursuit of automated testing on a project is still a logical fit, given the organizational needs

Benefits of Automated Testing

- Production of a reliable system
- Improvement of the quality of the test effort
- Reduction of the test effort and minimization of the schedule

Production of a Reliable System

- Improved performance testing
- Improved load/stress testing
- Quality measurements and test optimization
- Improved partnership with development team
- Improved system development life cycle

Improved Quality of the Test Effort - 1

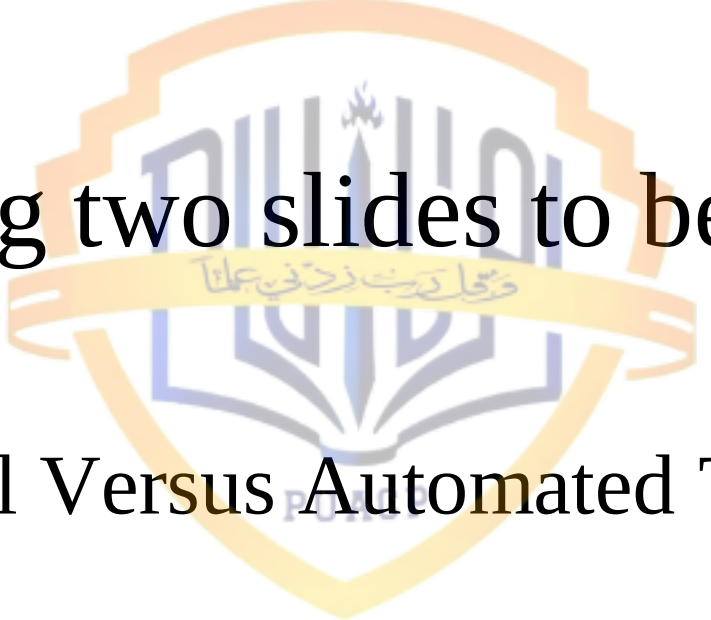
- Improved build verification testing
- Improved regression testing
- Improved multiplatform compatibility testing
- Improved software compatibility testing
- Improved execution of mundane tests

Improved Quality of the Test Effort - 2

- Improved focus on advanced test issues
- Execution of tests that manual testing can't accomplish
- Ability to reproduce software defects
- Documentation of business knowledge
- After-hour testing

Reduction of Test Effort and Minimization of Schedule

- Test plan development
- Test procedure development
- Test execution
- Test result analysis
- Error status/correction monitoring
- Report creation

The logo of the University of Al-Qadisiyah is a shield-shaped emblem. It features a stylized open book at the base, with a flame or torch rising from the center. Above the book, there is a banner with Arabic calligraphy. The entire emblem is surrounded by a decorative border. The text "Following two slides to be inserted" is overlaid on the upper part of the logo.

Following two slides to be inserted

Manual Versus Automated Testing

Manual Versus Automated Testing - 1

Test Steps	Manual Testing (Hrs)	Automated Testing (Hrs)	Percentage Improvement
Test Plan Development	32	40	-25%
Test Procedure Development	262	117	55%
Test Execution	466	23	95%
Test Result Analysis	117	58	50%

Manual Versus Automated Testing - 2

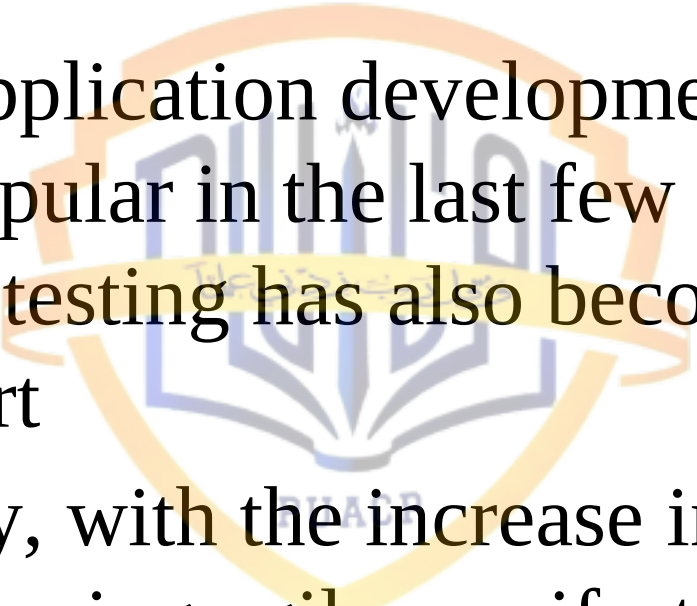
Test Steps	Manual Testing (Hrs)	Automated Testing (Hrs)	Percentage Improvement
Error Status / Correction Monitoring	117	23	80%
Report Creation	96	16	83%
Total Duration	1090	277	75%

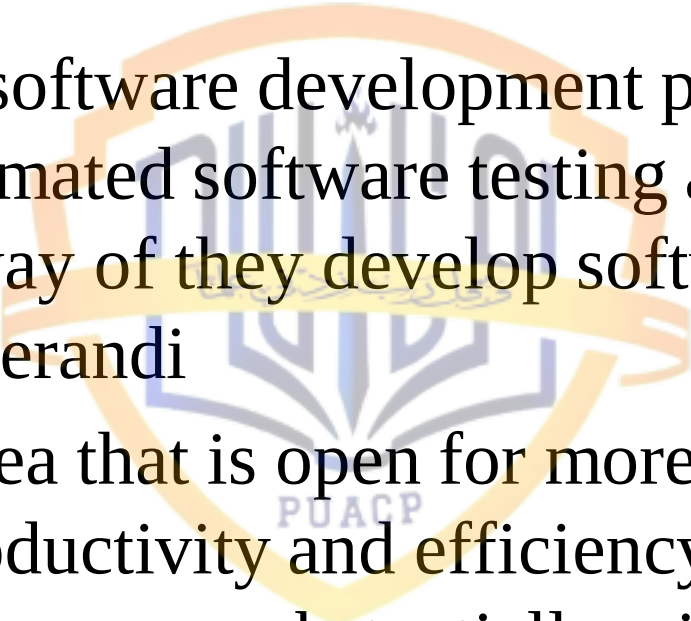
False Expectations for Automated Testing

- Automatic test plan generation
- Test tool fits all
- Immediate test effort reduction
- Immediate schedule reduction
- Tool ease of use
- Universal application of test automation
- One hundred percent coverage

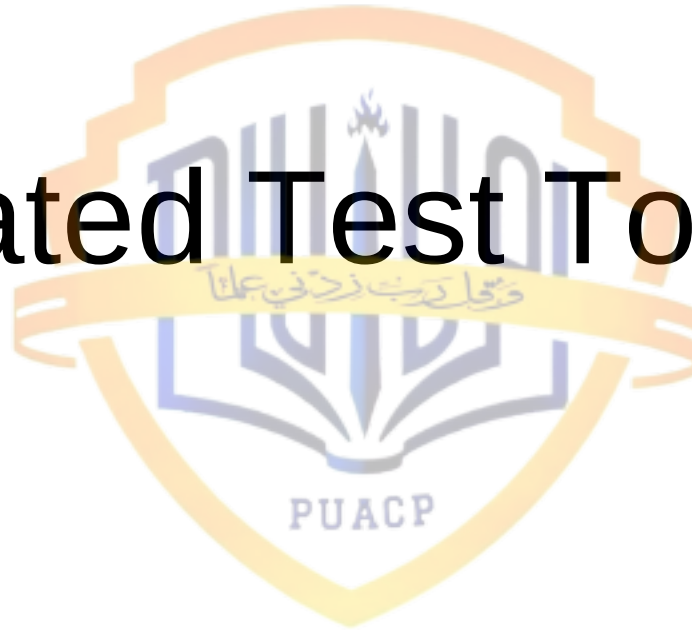
Automated Testing in Rapid Software Development



- 
- As rapid application development has become popular in the last few years, automated testing has also become an integral part
 - Particularly, with the increase in the number of projects using agile manifesto, automated software testing is being used extensively

- 
- Many agile software development processes include automated software testing as an integral part of the way of they develop software – a sort of *modus operandi*
 - This is an area that is open for more research and certainly productivity and efficiency of software testing can increase substantially with automated software testing

Automated Test Tools



Automated Test Tools - 1

- Usually support automated execution of black-box tests
- An automated test tool reads a test script, plays the script, and compare the actual results to the original or expected responses
- They enable repeatability and consistency of test execution as well as productivity

Automated Test Tools - 2

- Some tool suites also provide the ability to distribute or simulate the execution of tests over multiple processors, thus providing performance and stress tests to measure the system's ability to handle high volumes of transactions and users

Evaluation Criteria for Test Tools

- Compatibility
- Usability
- Maintainability
- Manageability

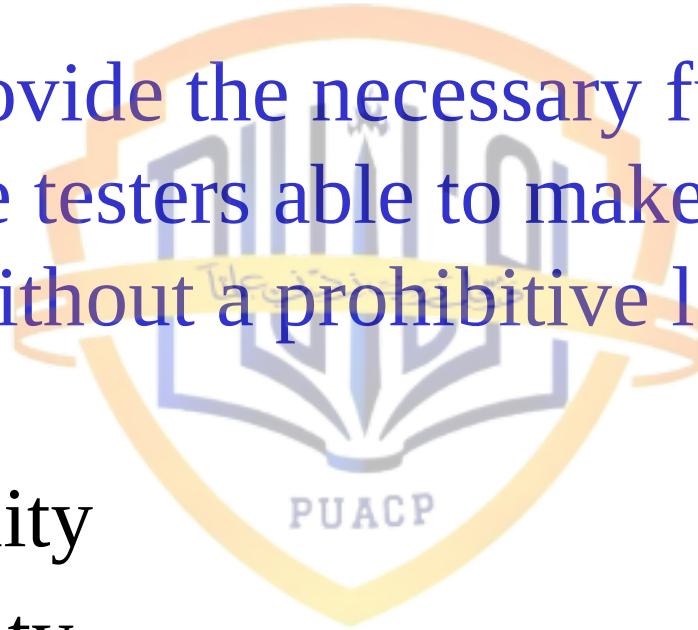


Compatibility

- Does the tool support the platforms and environments of the system under test?
- Client-Level Compatibility
- Communications-Level Compatibility
- Host-Level Compatibility
- Portability

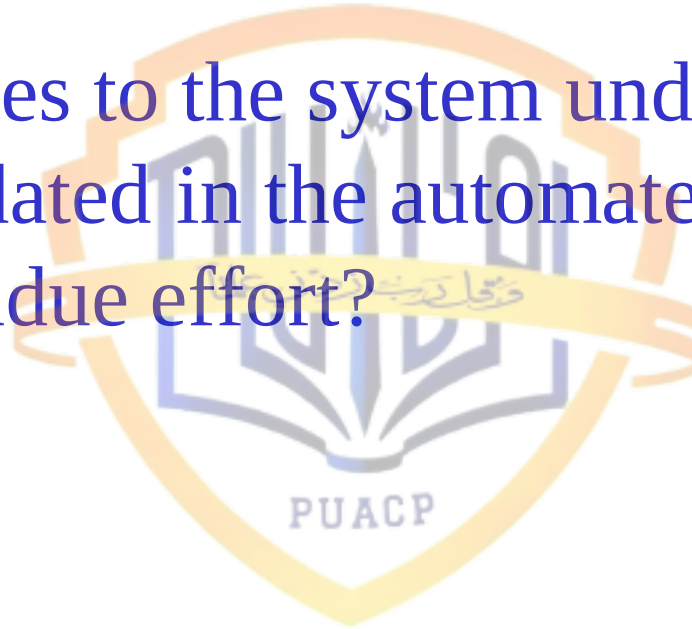
Usability

- Does it provide the necessary functionality, and are the testers able to make effective use of it without a prohibitive learning curve?
- Functionality
- Extensibility
- Learning Curve



Maintainability

- Can changes to the system under test be accommodated in the automated tests without undue effort?



Manageability

- Does the tool provide sufficient and meaningful information to enable management of the test library and measurement of the test results?
- Test Library Management

Test Results

- Test results should be meaningful to inform management of the status of the system under test and its quality relative to expectations
- Test tools should be reviewed to determine the amount and quality of test result information; analysis and reporting capabilities are either included or can be made available

Test Case Generation - 1

- Programmatic creation of test case data, as opposed to sampling or other techniques
- Advantage: High volumes of test conditions can be created
- Disadvantage: Volume alone is not guarantor of coverage or efficacy

Test Case Generation - 2

- Equivalence classes
 - Large number of test cases
 - More than one test case that exercises the same path, or too few to cover all the paths
- Cause-effect node graphing

Vendor Selection - 1

- Experience
 - What level of experience does the vendor offer in the implementation of automated test tools?
- Service
 - What services does the vendor offer for training as well as implementation to assure that the tool is properly deployed?

Vendor Selection - 2

- Support
 - How readily available and responsive is the vendor to technical questions and issues?
- Commitment
 - Is the vendor committed to the automated test tools market or is it only one of several offerings?

Summary

- We had a brief introduction of automated software testing
- In addition, we talked about the selection criteria for test tools from the technical aspect as well as from the aspect of vendors

References

- Automated Software Testing by Elfriede Dustin, Jeff Rashka, and John Paul, Addison-Wesley, 1999 (Chapters 1-3)
- Inroads to Software Quality by Alka Jarvis and Vern Crandall (Chapter 6.5)